Online Supplementary Appendix to "Improving the Simple Average Combined Forecast via Factor-Adjusted Regularization"

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The main paper treats the simple average as the sole common factor. In this Online Supplementary Appendix, we extend our framework to incorporate additional common factors beyond the simple average. This decomposition allows each individual forecast $f_{i,t+h}$ to be expressed as:

$$f_{i,t+h} = \bar{f}_{t+h} + \boldsymbol{\lambda}_i^{\prime} \boldsymbol{g}_{t+h} + d_{i,t+h}, \tag{A.1}$$

where $\boldsymbol{g}_{t+h}$ is an r-dimensional vector of additional common factors, and $\boldsymbol{\lambda}_i$ is the corresponding r-dimensional vector of loadings. Each forecast $f_{i,t+h}$ thus consists of $(r+1)$ common components, including the simple average $\bar{f}_{t+h}$.

In this context, the combined forecast is obtained by substituting Equation (A.1) into the linear combination:

$$\begin{aligned} f_{c,t+h} &= \sum_{i=1}^{N} \beta_i \left(\bar{f}_{t+h} + \boldsymbol{\lambda}_i^{\prime} \boldsymbol{g}_{t+h} + d_{i,t+h} \right) \\ &= \bar{f}_{t+h} + \boldsymbol{\alpha}^{\prime} \boldsymbol{g}_{t+h} + \sum_{i=1}^{N} \beta_i d_{i,t+h}, \end{aligned} \tag{A.2}$$

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where $\boldsymbol{\alpha} \equiv \sum_{i=1}^{N} \beta_i \boldsymbol{\lambda}_i^{'}$ is an r-dimensional vector representing the effect of the additional common factors $\boldsymbol{g}_{t+h}$ on the combined forecast. The contribution of the additional common factors $\boldsymbol{g}_{t+h}$ to the target variable y_{t+h} can be consistently quantified by regressing the deviation $(y_{t+h} - \bar{f}_{t+h})$ on $\boldsymbol{g}_{t+h}$. This is because the additional common factors $\boldsymbol{g}_{t+h}$ and the idiosyncratic components $d_{i,t+h}$ are uncorrelated.

We apply Principal Component Analysis (PCA) to estimate the additional common factors $(\boldsymbol{g}_{t+h})$, the corresponding loadings $(\boldsymbol{\lambda}_i)$, and the number of factors (r) following Stock and Watson (2002), Bai (2003), and Bai and Ng (2002). However, a key challenge is that these additional common factors may be "weak" after accounting for the simple average. To address this, we also consider Supervised PCA (SPCA) of Giglio et al. (2025) as an alternative to standard PCA. Unlike the Ridge regression, which we use to handle potentially weak idiosyncratic components $(d_{i,t+h})$ in the main paper, SPCA is employed here to address potentially weak common components $(\boldsymbol{g}_{t+h})$ by selecting a subset of forecasts to strengthen the extracted factors.

The SPCA procedure is iterative, combining supervised selection and factor extraction. It begins by computing the univariate correlation between each $d_{i,t+h}$ (where $d_{i,t+h} = f_{i,t+h} - \bar{f}_{t+h}$ from the previous subsection) and the target variable y_{t+h}. Components $d_{i,t+h}$ with sufficiently high absolute correlation are selected. The first estimated factor is then constructed using PCA applied to these selected components. These two steps are repeated r times. In each subsequent iteration, the selection and extraction are performed on the residuals of y_{t+h} and $d_{i,t+h}$, both obtained after removing the portion explained by the common factors estimated in the previous iteration. The number of additional common factors, r, is determined using 3-fold cross-validation to minimize the MSFE. After r iterations, the residuals of $d_{i,t+h}$ are purely idiosyncratic components.

We rearrange Equation (A.2) and rewrite it in terms of forecast errors as follows:

$$\tilde{u}_{t+h} = \sum_{i=1}^{N} \beta_i d_{i,t+h} + u_{c,t+h}, \tag{A.3}$$

where $\tilde{u}_{t+h} \equiv y_{t+h} - \left(\bar{f}_{t+h} + \boldsymbol{\alpha}'\boldsymbol{g}_{t+h}\right)$ represents the forecast error after removing all common components, including the simple average, and $u_{c,t+h}$ is the combined forecast error. We estimate β_i by regressing the forecast errors $\tilde{u}_{t+h}$ on the idiosyncratic components $d_{i,t+h}$.

This structure addresses the limitation of the Diebold and Shin (2019). While Diebold and Shin (2019) also use a forecast encompassing approach, their reliance on sparsity in the coefficients δ_i can be problematic when individual forecasts contain additional common factors beyond the simple average, leading to high correlation among the individual forecasts $f_{i,t+h}$. In contrast, our method explicitly accounts for these additional common factors, thereby allowing for sparsity in the true weights β_i because the resulting idiosyncratic components ($d_{i,t+h}$) are weakly correlated.

To evaluate the performance of this extended framework, we apply it to the empirical applications presented in Section 4 of the main paper. Table A1 reports the results for the case involving common factors beyond the simple average. FARM improves the simple average combined forecast by incorporating selected idiosyncratic components, as shown in the results presented in Table 1 of the main paper. FARM2 and FARM3 further enhance the simple average forecast by selectively incorporating both additional common factors and idiosyncratic components. In particular, FARM2 uses PCA to estimate the additional common factors, while FARM3 employs SPCA. We set the maximum number of additional common components, r, to 2 for these empirical applications.

Interestingly, incorporating additional common components does not further improve the simple average forecast. This finding suggests that FARM1 ($r = 0$) may be sufficient, implying that FARM2 or FARM3 may be unnecessary in these particular applications. One explanation is that the simple average already captures the most important common factor shared across individual forecasts, while estimating additional common factors and their corresponding coefficients ($\boldsymbol{\alpha}$) may introduce estimation errors that offset any potential gains. However, a comparison between FARM2 and FARM3 shows that FARM3 often outperforms FARM2. This result highlights that any additional common factors beyond the simple average in these applications are likely weak.

Table A1: MSFEs Relative to the Simple Average Combined Forecast

		RPI	CPIAUCSL	PCEPI	INDPRO	UNRATE	RPI	CPIAUCSL	PCEPI	INDPRO	UNRATE	RPI	CPIAUCSL	PCEPI	INDPRO	UNRATE
				$h=1$					$h=2$					$h=3$		
FARM	Lasso	1.034 (0.689)	0.990 (0.464)	0.827 **(0.038)**	0.998 (0.488)	0.931 (0.193)	1.052 (0.886)	1.005 (0.518)	1.117 (0.775)	1.154 (0.927)	0.914 (0.165)	1.071 (0.956)	1.133 (0.894)	1.020 (0.650)	1.035 (0.695)	0.961 (0.217)
	ALasso	0.966 (0.327)	0.885 **(0.077)**	0.902 **(0.024)**	0.935 (0.151)	0.862 **(0.054)**	0.986 (0.123)	0.962 (0.157)	1.024 (0.683)	0.985 (0.356)	0.859 **(0.027)**	1.015 (0.806)	1.085 (0.827)	1.018 (0.633)	0.985 (0.298)	1.005 (0.539)
	L_2-Boosting	0.961 (0.241)	0.883 **(0.004)**	0.866 **(0.002)**	0.985 (0.340)	0.877 **(0.019)**	1.014 (0.697)	0.929 **(0.070)**	0.932 **(0.031)**	1.086 (0.972)	0.877 **(0.021)**	1.000 (0.504)	1.005 (0.557)	1.014 (0.631)	0.961 (0.114)	0.948 **(0.086)**
	Ridge	1.007 (0.829)	0.932 **(0.053)**	0.913 **(0.007)**	1.016 (0.946)	0.926 **(0.082)**	0.998 (0.395)	0.997 **(0.043)**	0.952 **(0.027)**	1.005 (0.577)	0.927 **(0.050)**	1.006 (0.797)	1.001 (0.926)	1.003 (0.997)	1.013 (0.913)	0.969 **(0.033)**
FARM2	Lasso	0.968 (0.375)	0.983 (0.434)	0.900 **(0.071)**	1.214 (0.993)	1.123 (0.991)	1.024 (0.859)	1.153 (0.869)	1.167 (0.815)	1.186 (0.964)	1.051 (0.853)	1.048 (0.885)	1.154 (0.897)	1.020 (0.671)	1.132 (0.964)	1.162 (1.000)
	ALasso	0.965 (0.264)	0.892 **(0.050)**	0.931 **(0.039)**	1.117 (0.963)	1.053 (0.950)	0.997 (0.348)	0.960 (0.194)	0.981 (0.321)	1.159 (0.951)	1.069 (0.982)	1.014 (0.743)	1.070 (0.855)	1.006 (0.552)	1.037 (0.737)	1.099 (1.000)
	L_2-Boosting	0.954 (0.168)	0.902 **(0.002)**	0.911 **(0.002)**	1.078 (0.985)	1.081 (1.000)	1.010 (0.748)	0.970 (0.207)	0.969 (0.135)	1.144 (0.988)	1.055 (0.979)	0.999 (0.481)	1.020 (0.799)	1.027 (0.786)	1.019 (0.671)	1.137 (1.000)
	Ridge	1.005 (0.727)	0.952 **(0.021)**	0.993 (0.167)	1.063 (1.000)	1.102 (1.000)	1.005 (0.734)	1.001 (0.677)	1.002 (0.707)	1.070 (0.967)	1.095 (1.000)	1.003 (0.717)	1.003 (0.752)	1.003 (0.743)	1.065 (0.995)	1.105 (1.000)
FARM3	Lasso	1.008 (0.546)	1.119 (0.868)	0.916 **(0.076)**	1.028 (0.699)	1.046 (0.824)	1.010 (0.662)	1.070 (0.778)	1.103 (0.815)	1.160 (0.948)	0.938 (0.162)	1.036 (0.872)	1.132 (0.868)	1.010 (0.583)	1.010 (0.567)	1.000 (0.497)
	ALasso	0.971 (0.350)	0.969 (0.213)	0.867 **(0.004)**	0.971 (0.262)	1.024 (0.765)	1.024 (0.879)	0.951 (0.135)	1.037 (0.808)	0.983 (0.327)	0.925 **(0.074)**	1.029 (0.893)	1.074 (0.798)	1.015 (0.623)	0.996 (0.438)	1.010 (0.623)
	L_2-Boosting	0.957 (0.207)	0.935 **(0.023)**	0.928 **(0.007)**	0.998 (0.462)	1.009 (0.647)	1.017 (0.807)	0.949 **(0.071)**	0.963 **(0.052)**	1.090 (0.989)	0.934 **0.046**	0.998 (0.406)	1.003 (0.542)	1.010 (0.600)	0.973 (0.149)	0.977 (0.189)
	Ridge	1.006 (0.814)	0.947 **(0.041)**	0.938 **(0.008)**	1.034 (0.998)	1.016 (0.775)	1.000 (0.519)	0.999 (0.171)	0.977 **(0.078)**	1.023 (0.857)	0.963 **(0.087)**	1.006 (0.815)	1.001 (0.909)	1.002 (0.967)	1.026 (0.991)	0.988 (0.171)

[1] Table A1 reports the h-month-ahead out-of-sample forecasting performance for five U.S. macroeconomic series relative to the simple average combined forecast, where $h = 1, 2, 3$. A relative MSFE below one indicates that the combined forecast from a given method outperforms the simple average; such values are highlighted in blue. The p-value reported in parentheses is based on the Diebold and Mariano (1995) test. Values in bold indicate significance at the 10% level.

References

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